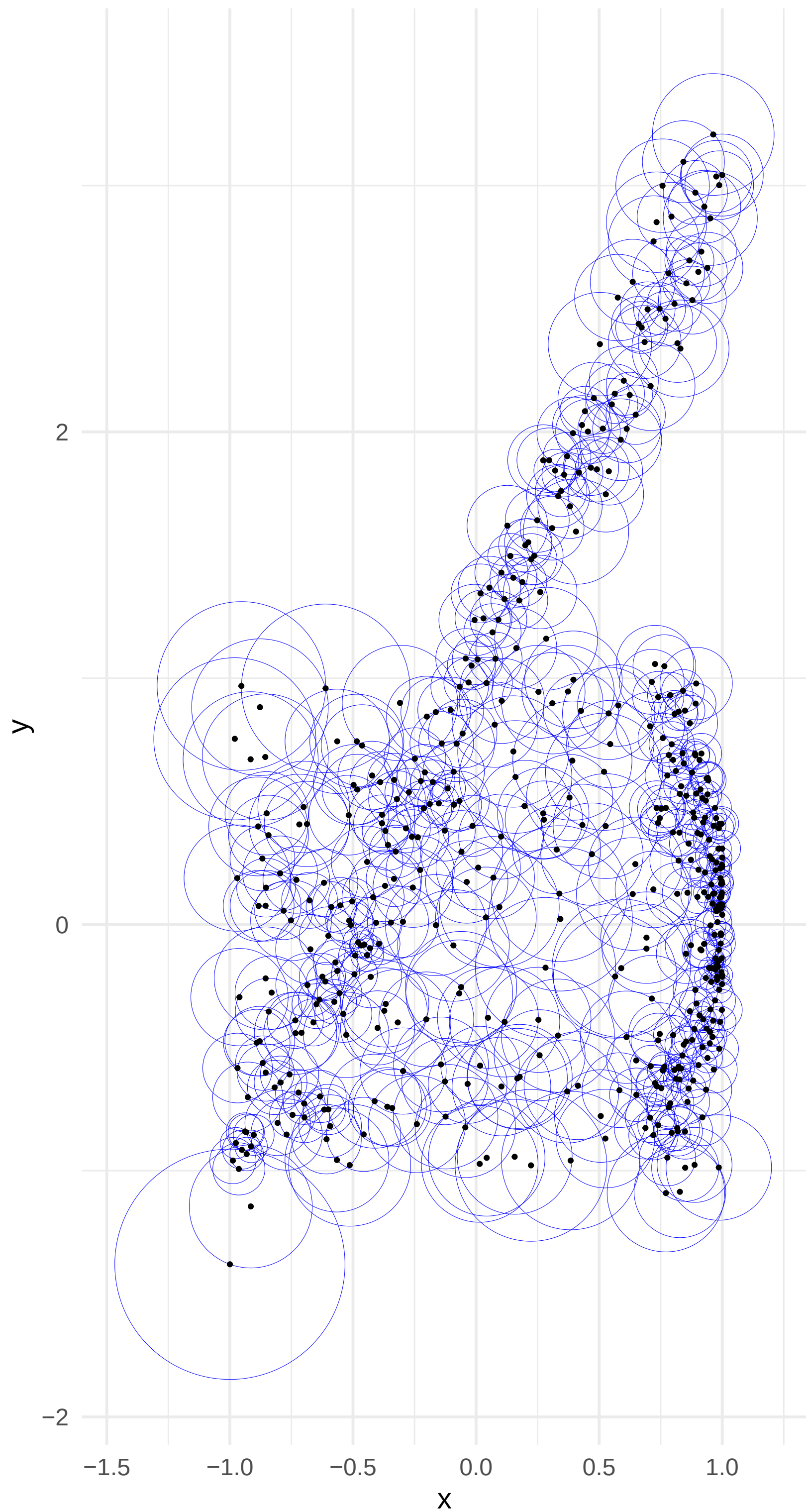
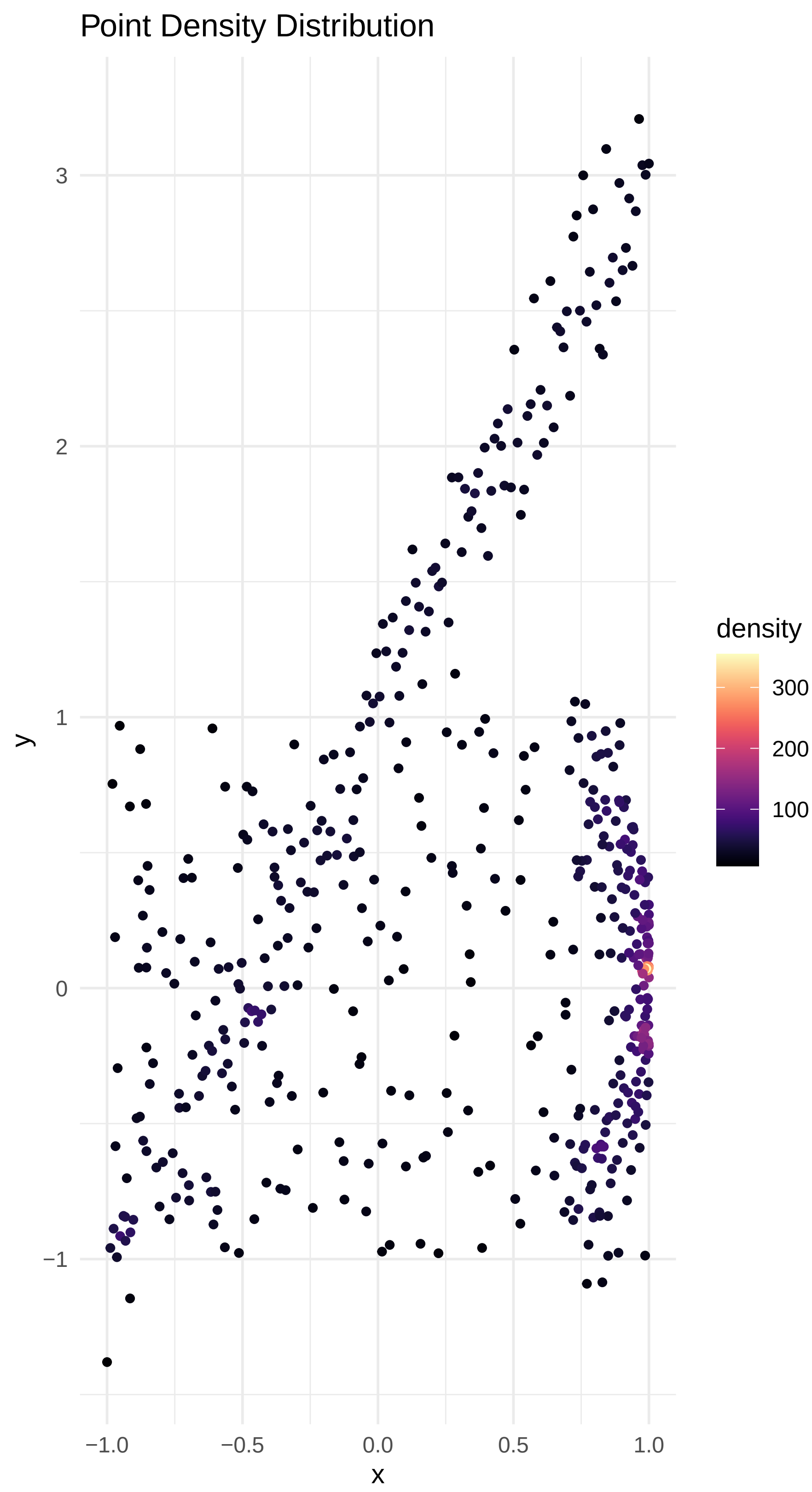
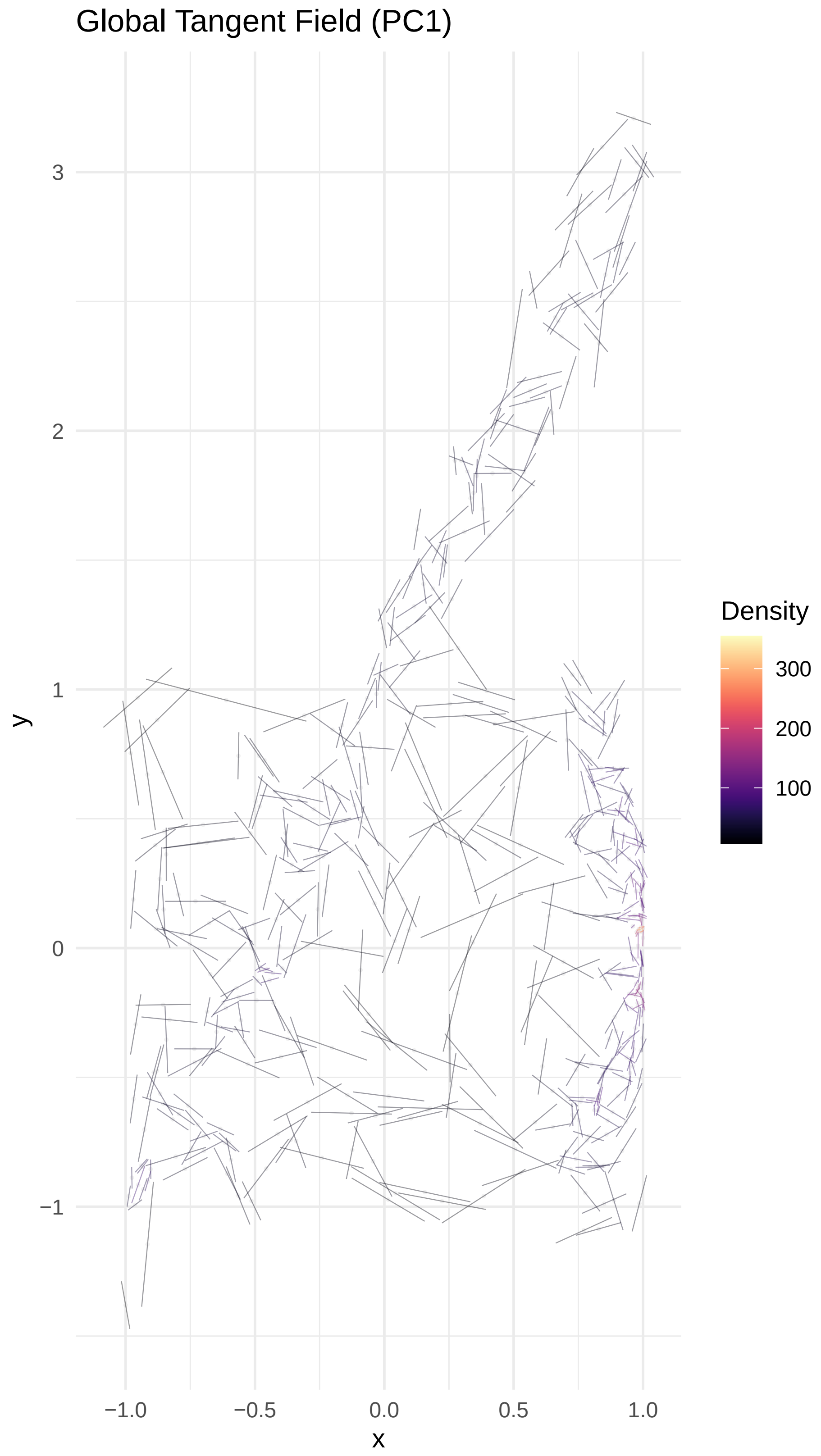
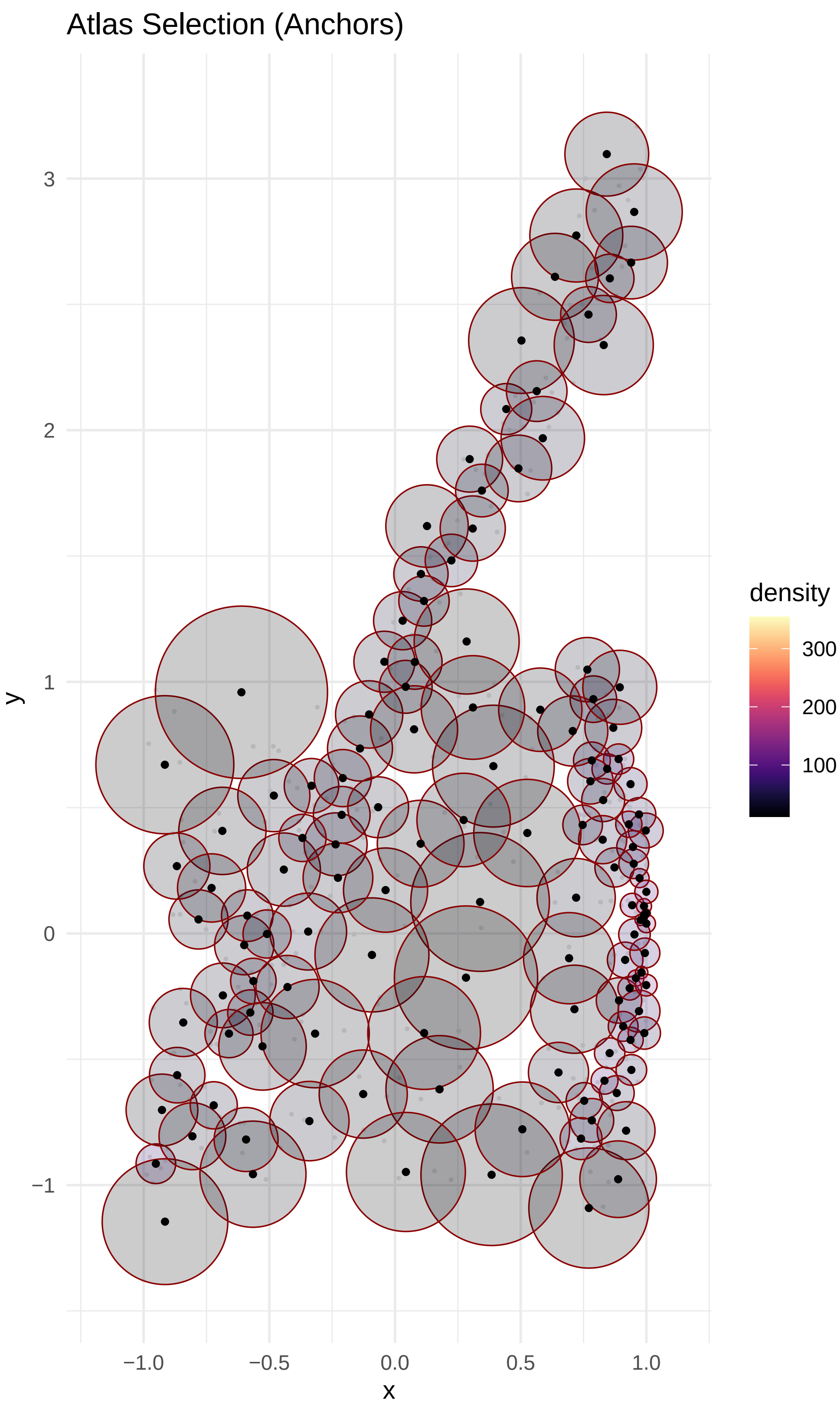
All Adaptive Spheres

Visualizing local neighborhood size (radius) for every point.




Point Density Distribution
density
300
200
100
y
3
2
1
0
−1
x
−1.0
−0.5
0.0
0.5
1.0

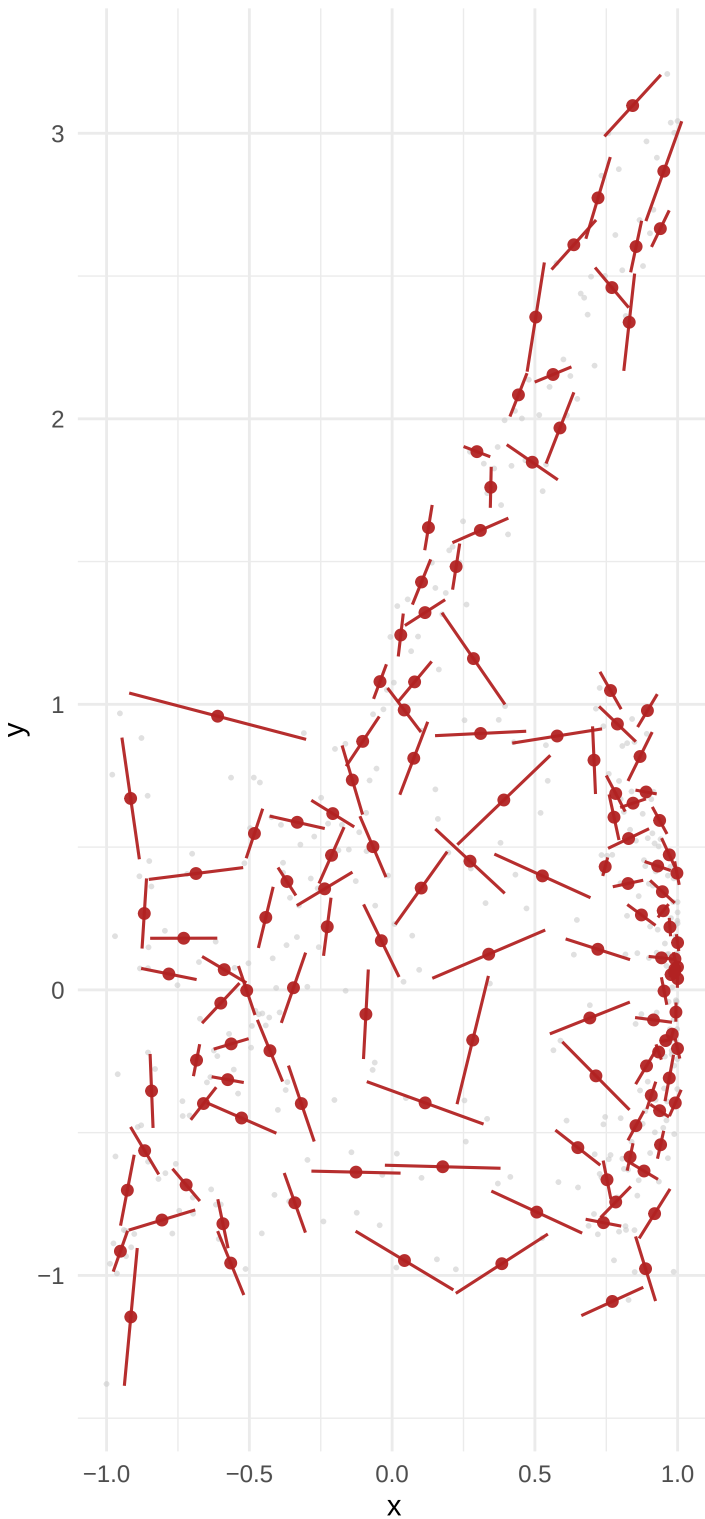

Global Tangent Field (PC1)
y
3
2
1
0
−1
x
−1.0
−0.5
0.0
0.5
1.0
Density
300
200
100



Atlas Selection (Anchors)
y
3
2
1
0
−1
x
−1.0
−0.5
0.0
0.5
1.0
density
300
200
100

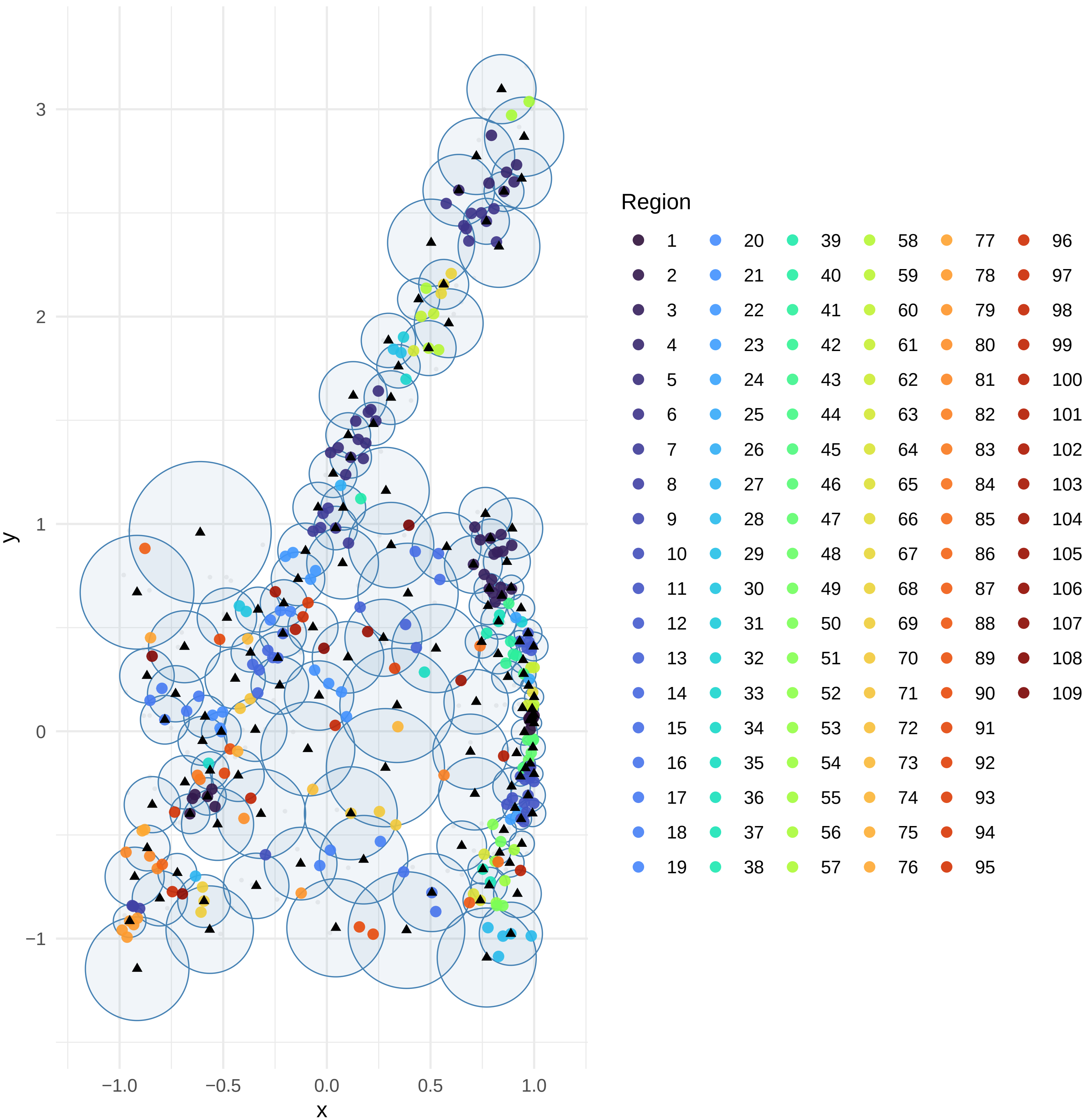
Anchor SVD Segments

Same anchors as plot 4: spheres first, then their SVD (PC1) in red



Intersection Regions (2+ anchors)

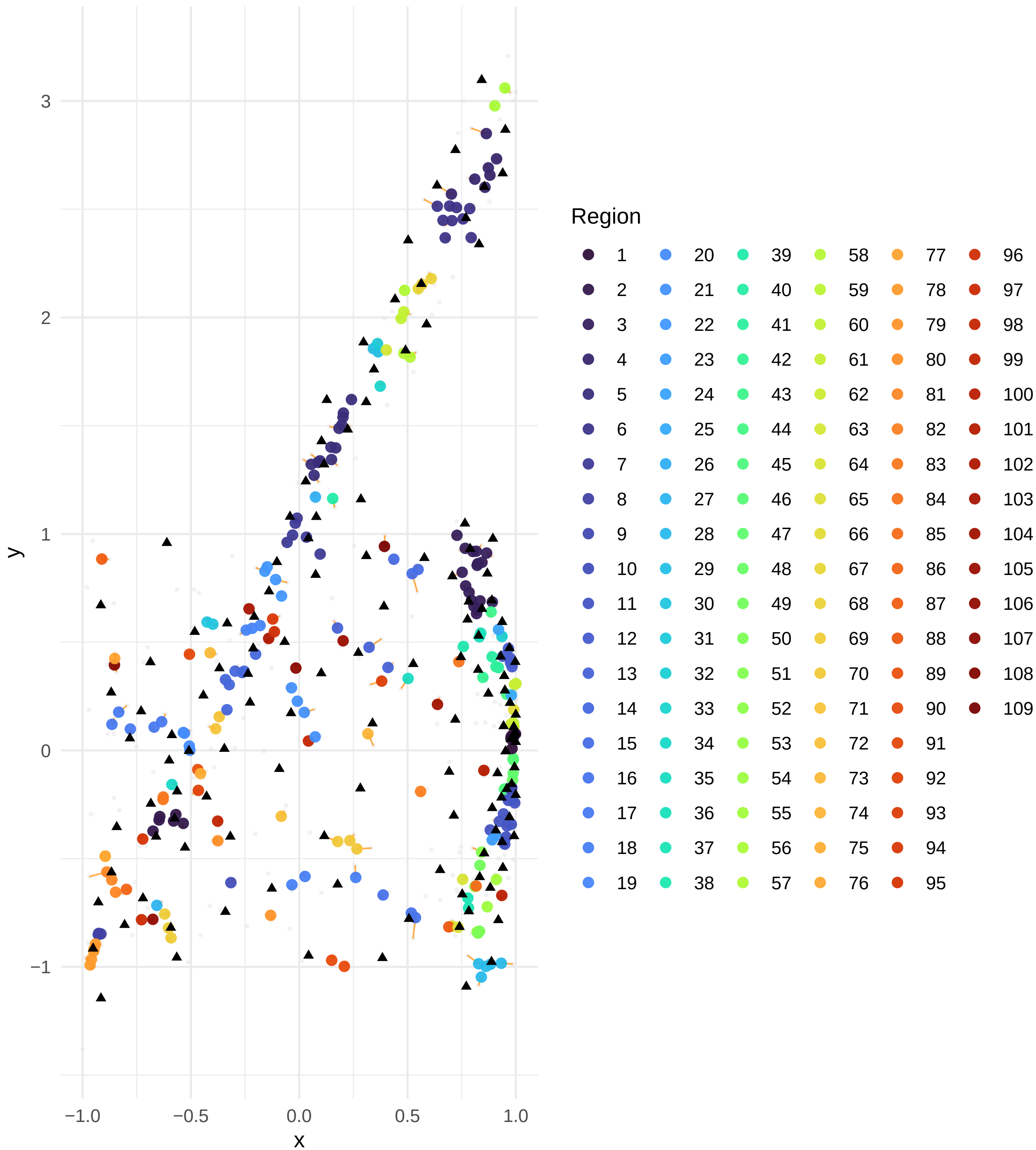
Anchor spheres in blue. Colored points = overlap zones. Triangles = anchor centers.

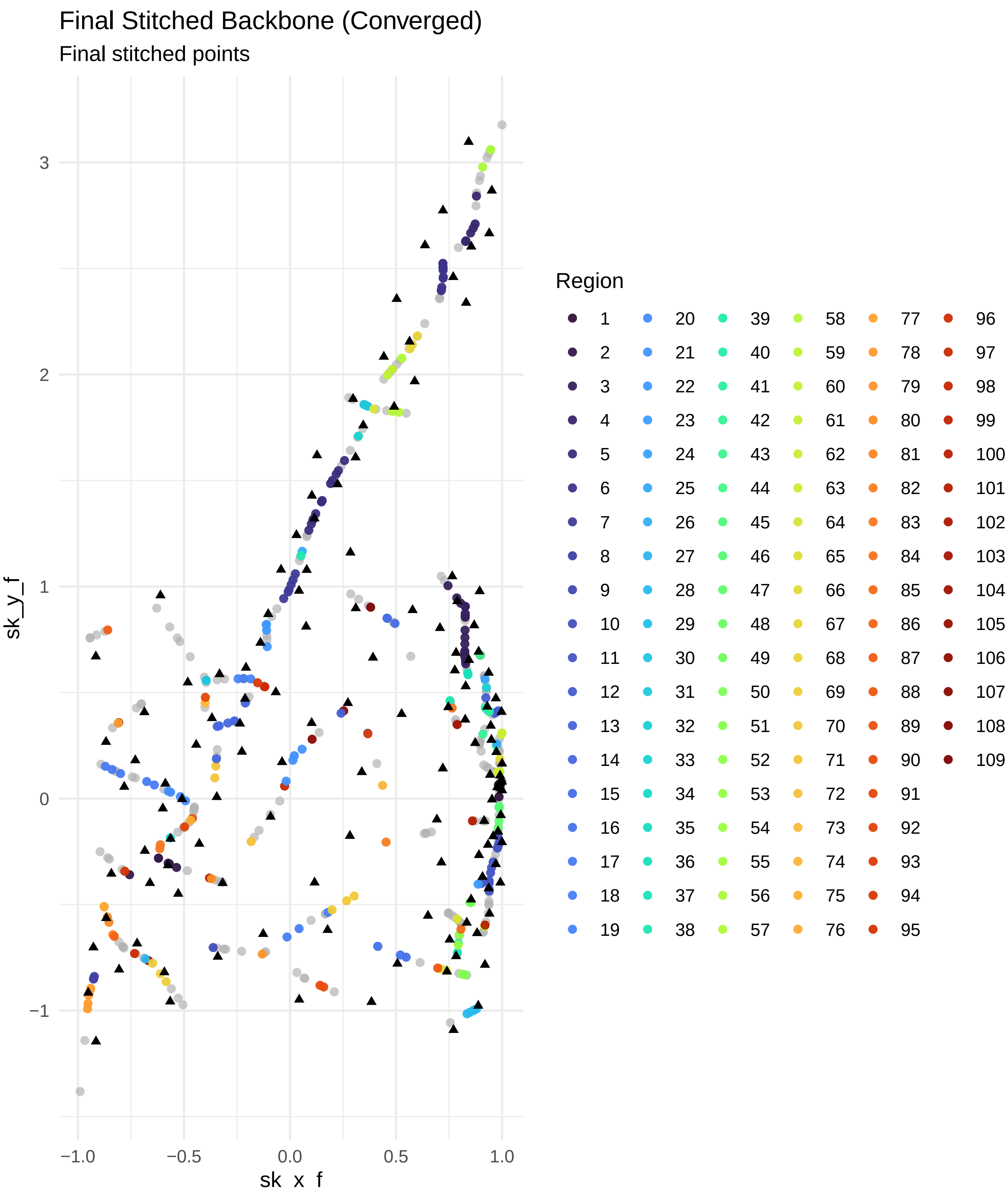


hold = 3.0, o_max = 0.3, o_min = 0.10, stability = 0.90, scale_factor = 2.5, max_iter = 50, conv_tol = 0.01

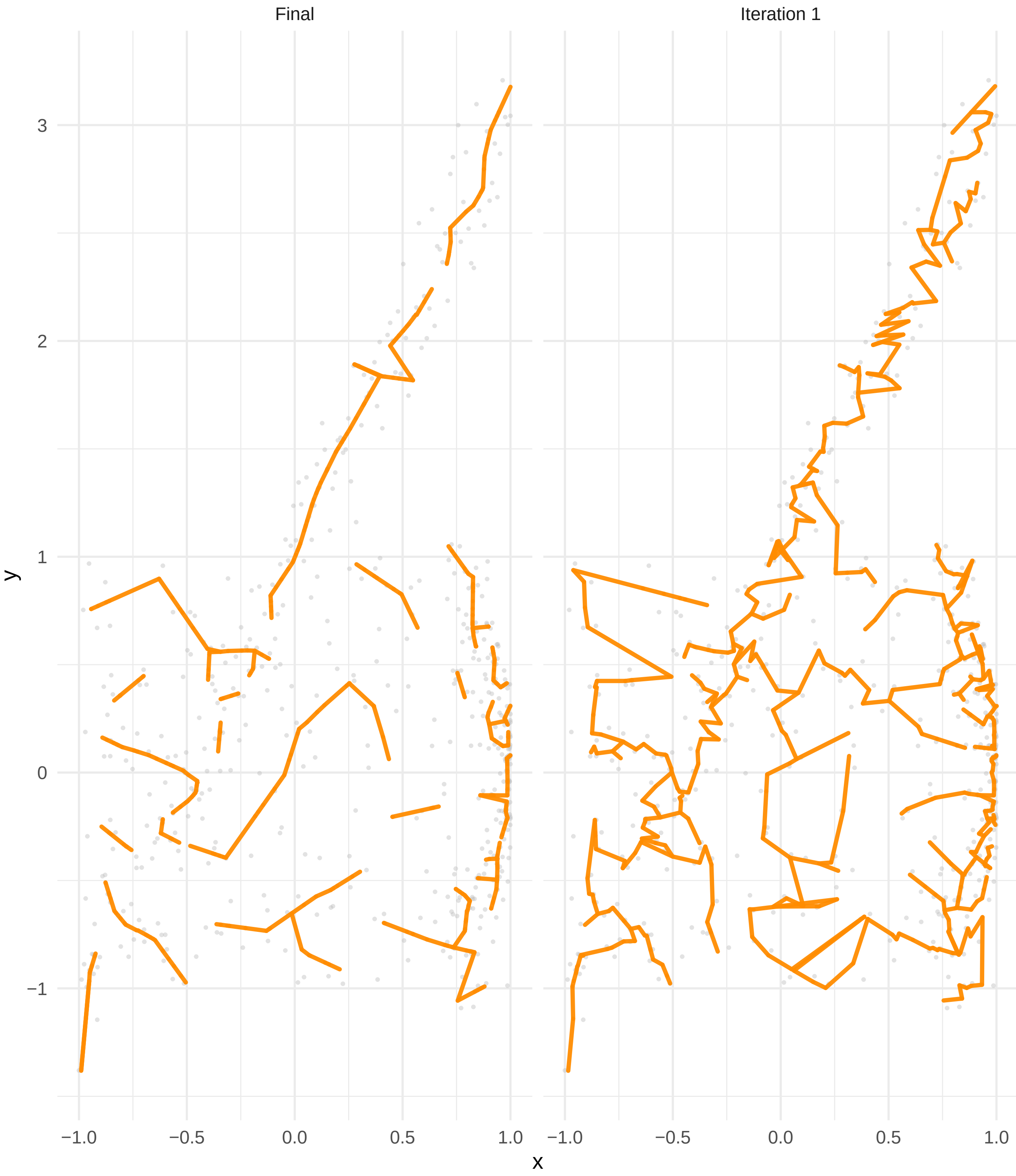
Stitched Backbone (Iteration 1)

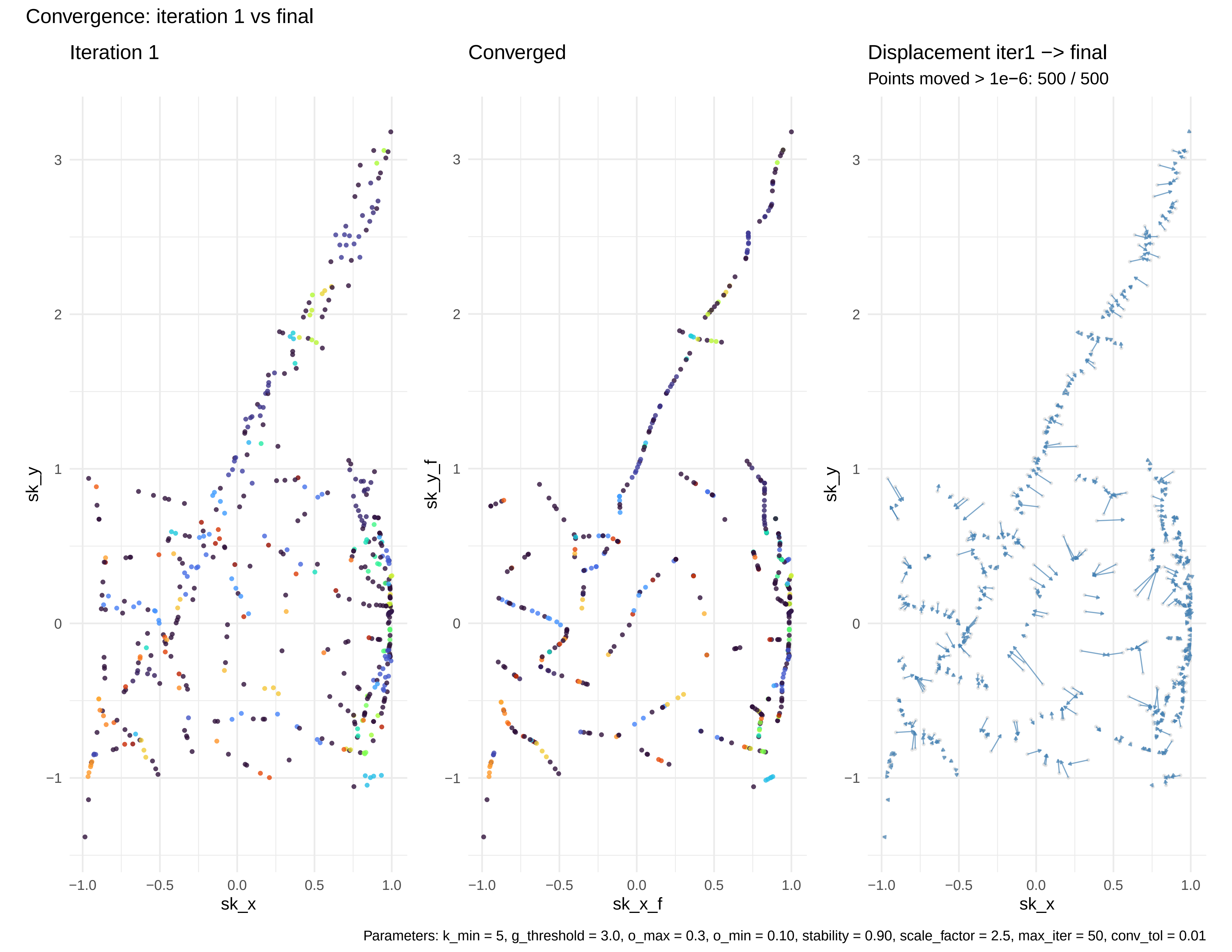
Orange path: stitching; Triangles: original anchors




Final Stitched Backbone (Converged)
Final stitched points
sk_y_f
sk_x_f
3
2
1
0
−1
−1.0
−0.5
0.0
0.5
1.0
Region
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38
39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57
58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76
77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95
96 97 98 99 100 101 102 103 104 105 106 107 108 109

Backbone Edges





Convergence: iteration 1 vs final
Iteration 1
Converged
Displacement iter1 −> final
Points moved > 1e−6: 500 / 500
sk_y
sk_x
sk_y_f
sk_x_f
3
2
1
0
−1
−1.0
−0.5
0.0
0.5
1.0
Parameters: k_min = 5, g_threshold = 3.0, o_max = 0.3, o_min = 0.10, stability = 0.90, scale_factor = 2.5, max_iter = 50, conv_tol = 0.01